

LARGE SCALE DIGITAL DRIVING OF PROGRAMMABLE PHOTONIC CHIPS

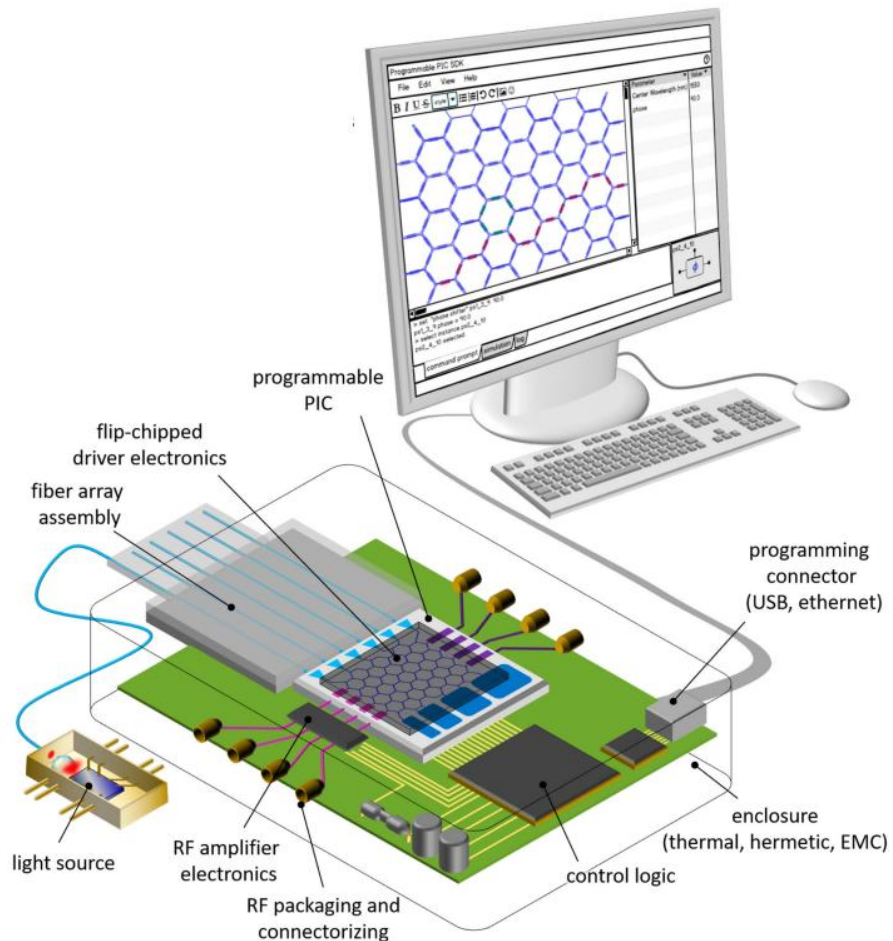
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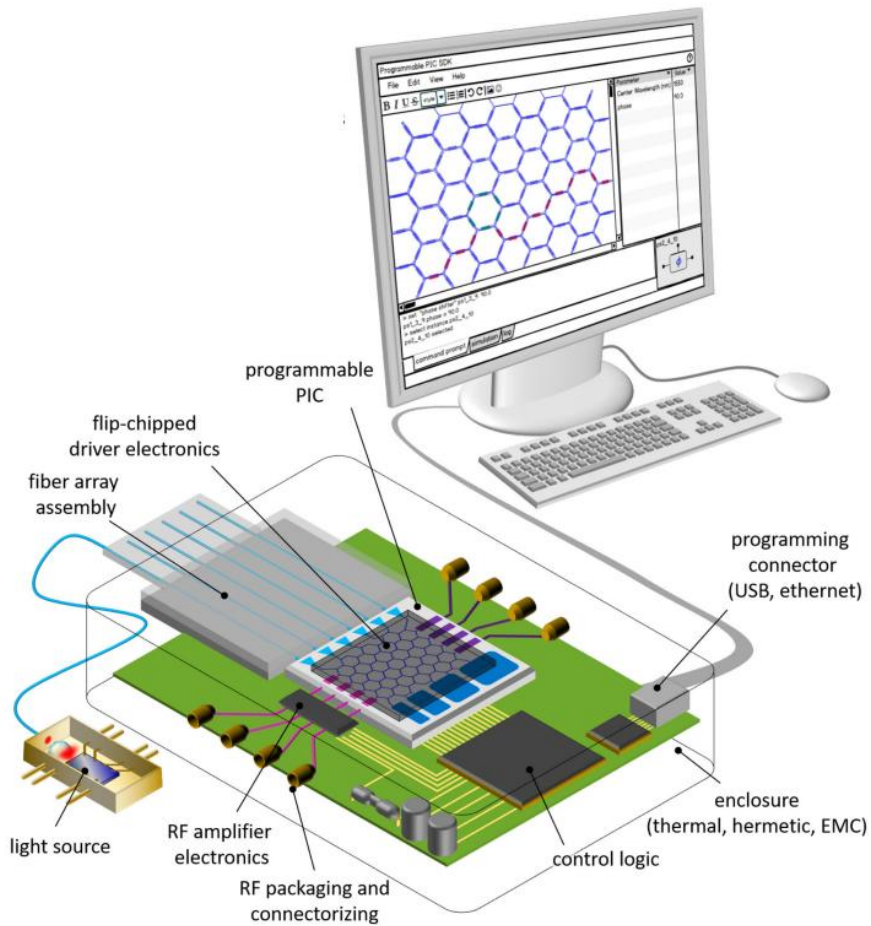
Sep 5th, 2022

Thesis Story



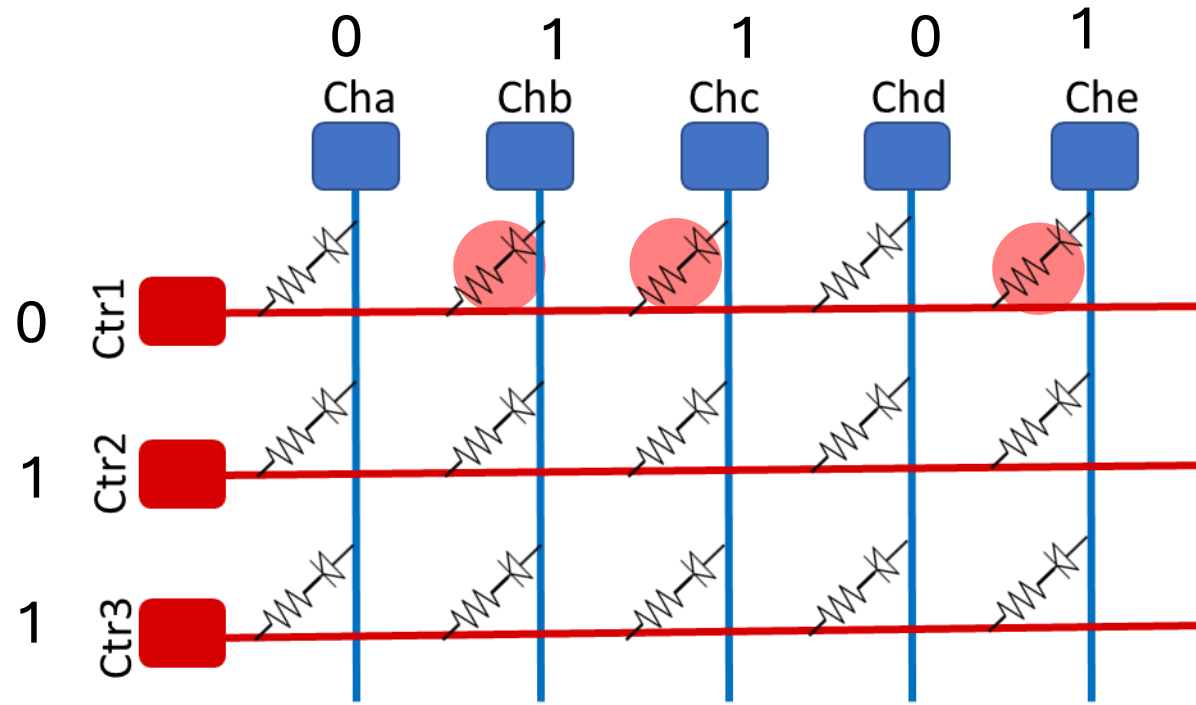
- Photonics are more electronically controlled.
- How to drive and control all the electronics.
 - Digital Driving.
 - Matrix addressing configuration.

Overview

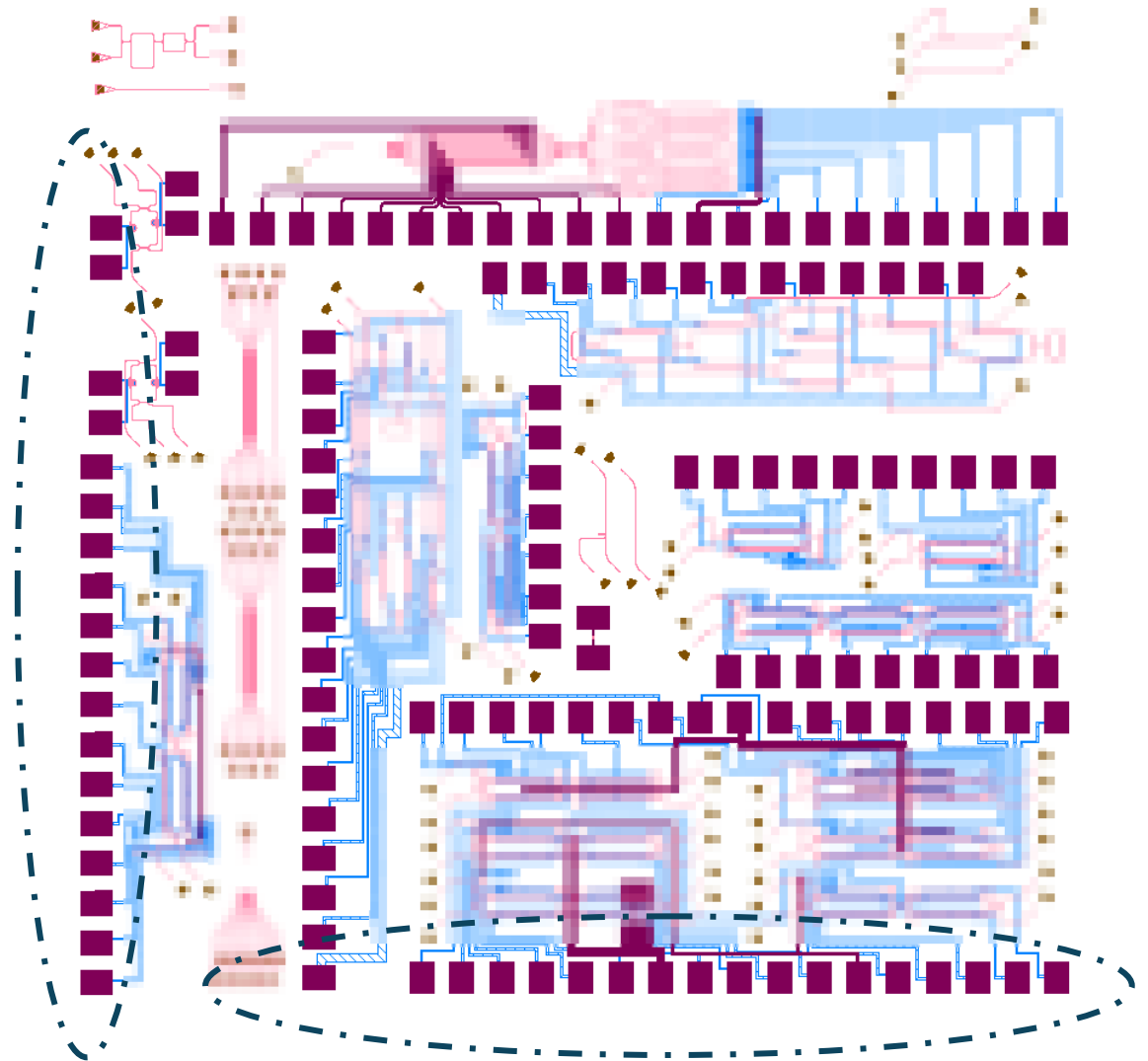


- Programmable photonics circuits.
- Mach Zehnder Interferometry (MZI) is main building unit.
- Light controlled by Optical Phase Shifter heaters.

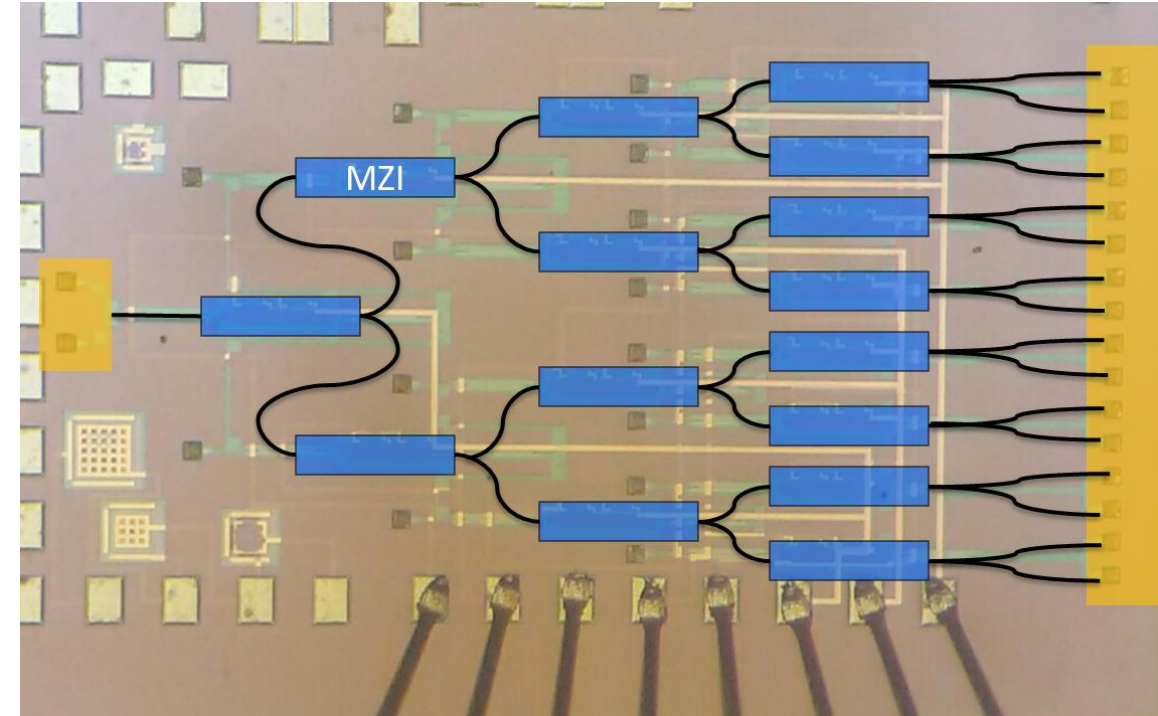
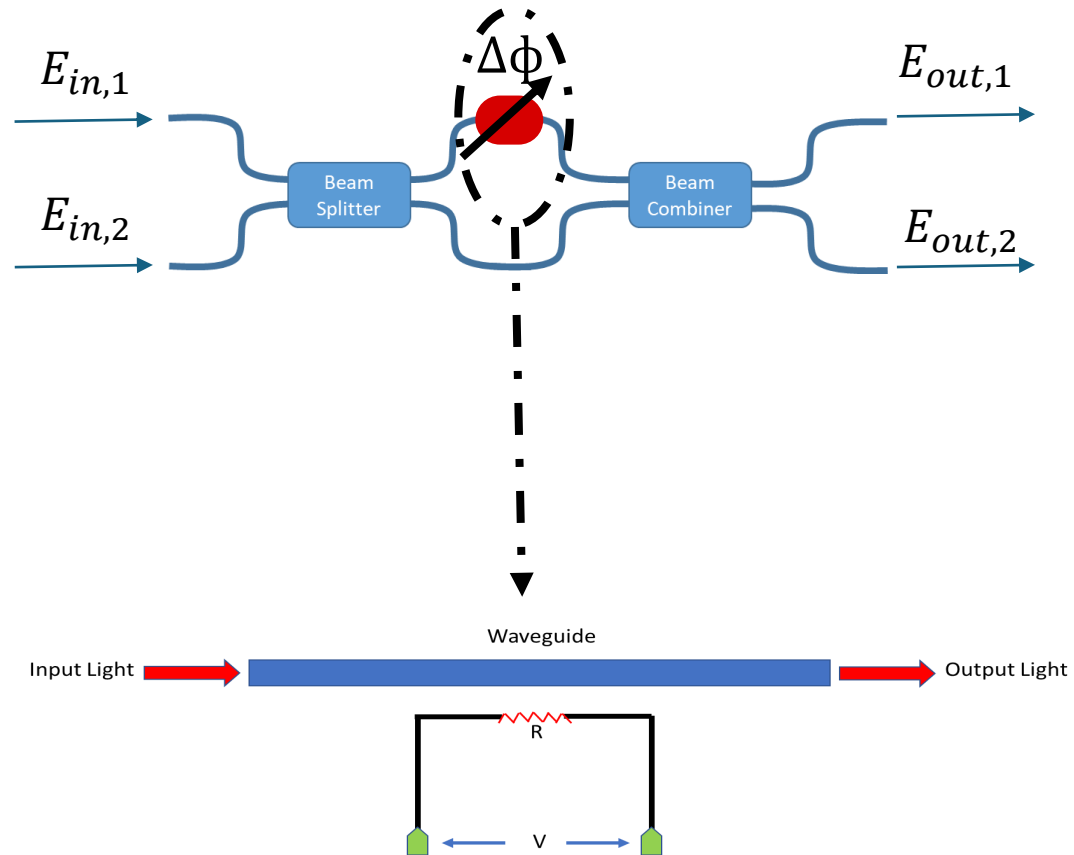
Overview



Only $N + M$ pads needed instead of $N \times M$

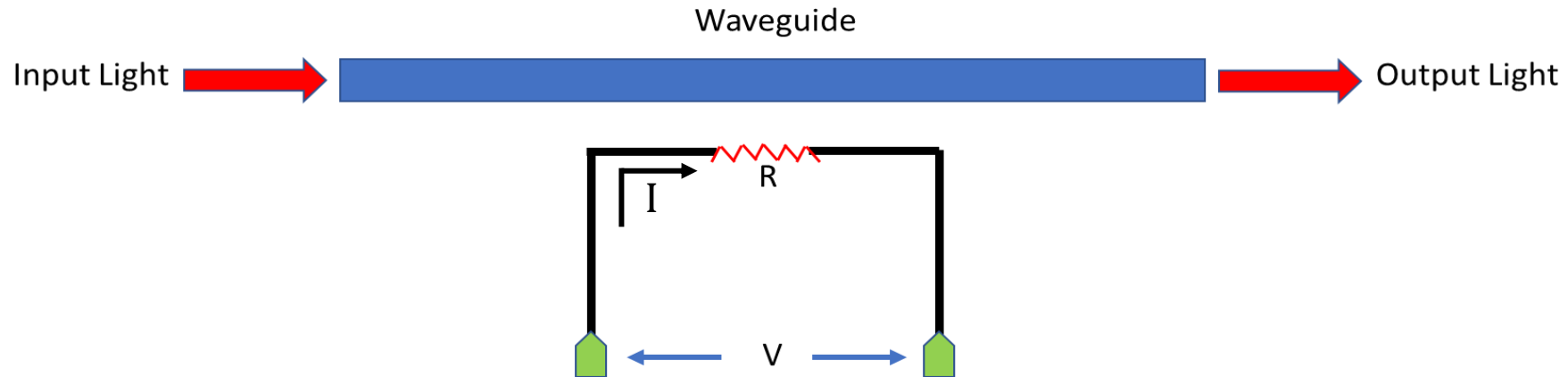


Mach Zehnder Interferometry (MZI)



Light controlled by Optical Phase Shifter heaters.

Phase Shifter Heaters

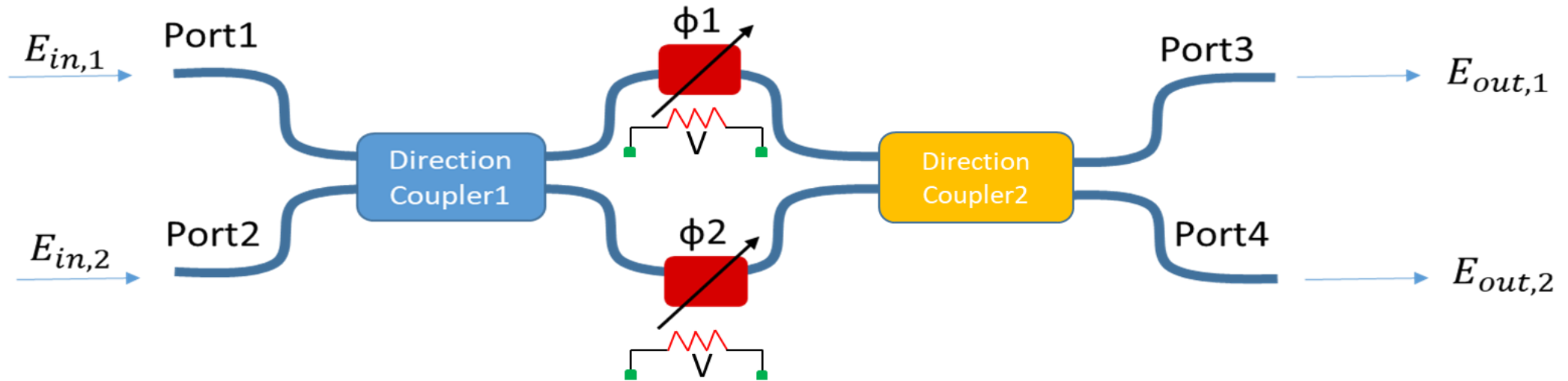


$$P \propto (V^2 / R) \propto \Delta T \propto \Delta n \propto \Delta \phi$$

Silicon thermo-optic coefficient = $dn/dT = 1.8 * 10^{-4} K^{-1}$

$$\Delta \phi = \Delta \beta * L = \frac{2\pi \Delta n * L}{\lambda}$$

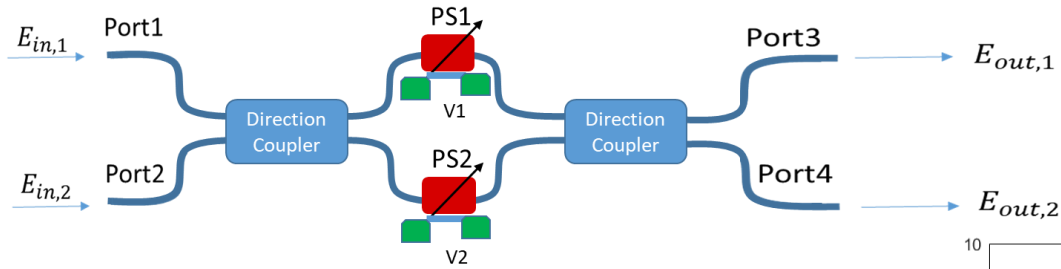
Mach-Zehnder Interferometer Phase Tuning



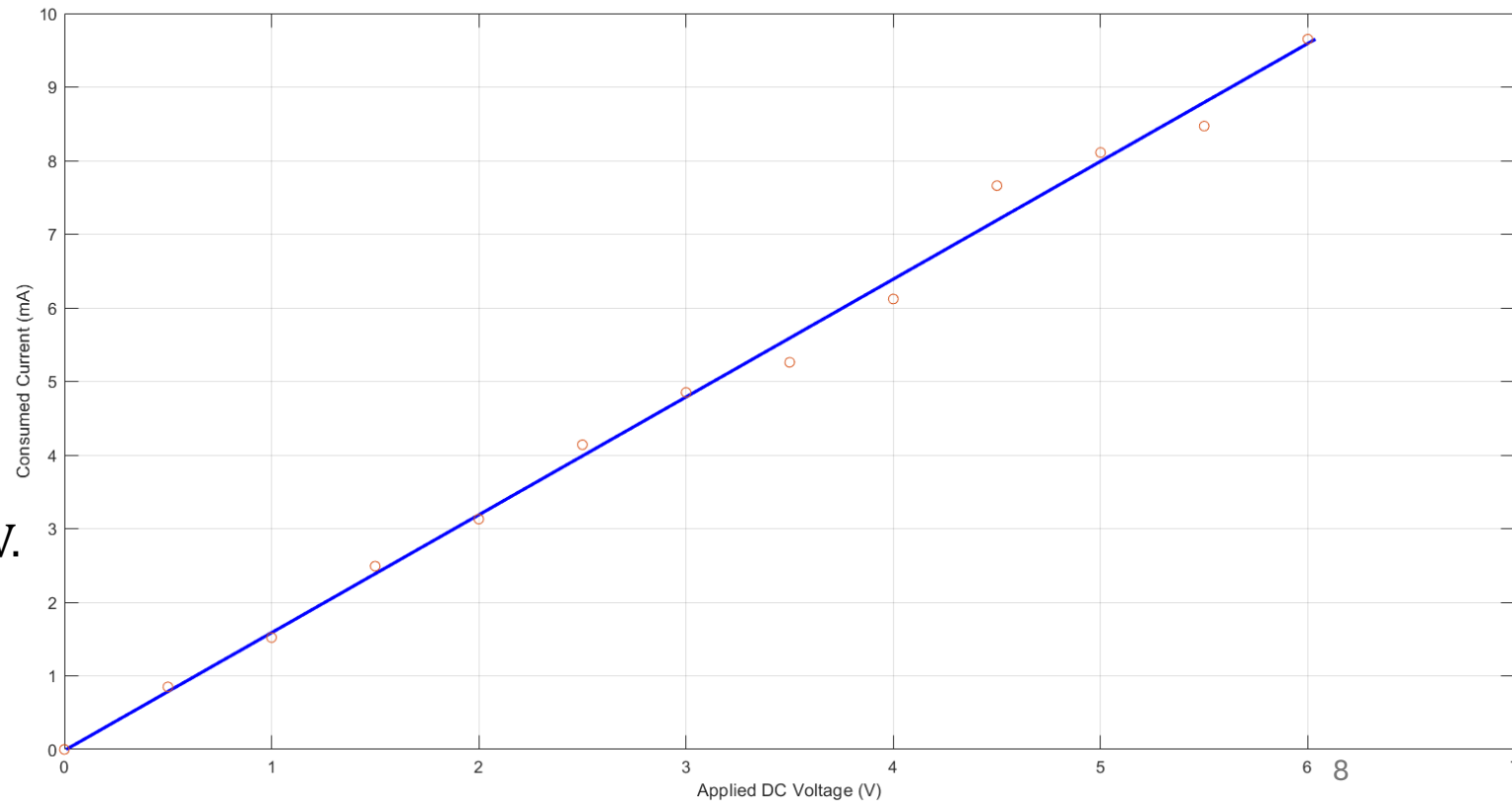
$$\begin{bmatrix} E_{out,1} \\ E_{out,2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -j\frac{1}{\sqrt{2}} \\ -j\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} e^{-j\phi_1} & 0 \\ 0 & e^{-j\phi_2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -j\frac{1}{\sqrt{2}} \\ -j\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} E_{in,1} \\ E_{in,2} \end{bmatrix}$$

$$\left| \frac{E_{out,1}}{E_{in,1}} \right|^2 = \sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) \quad \& \quad \left| \frac{E_{out,2}}{E_{in,1}} \right|^2 = \cos^2 \left(\frac{\phi_2 - \phi_1}{2} \right)$$

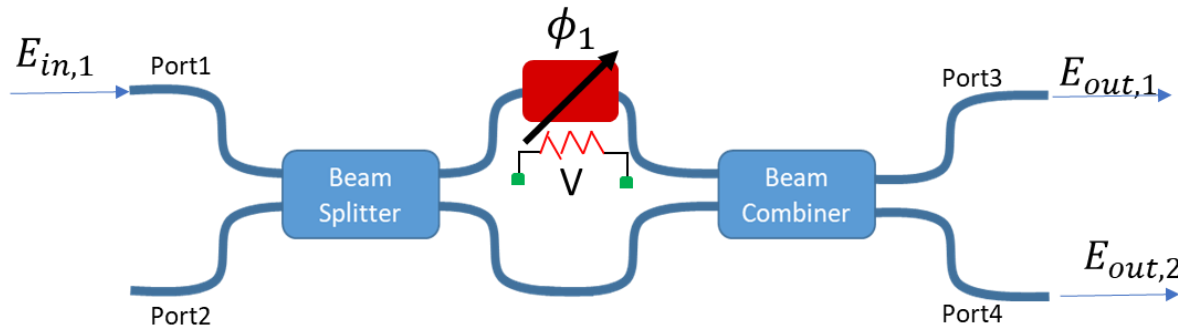
Digital driving for the Optical Phase Shift Heater



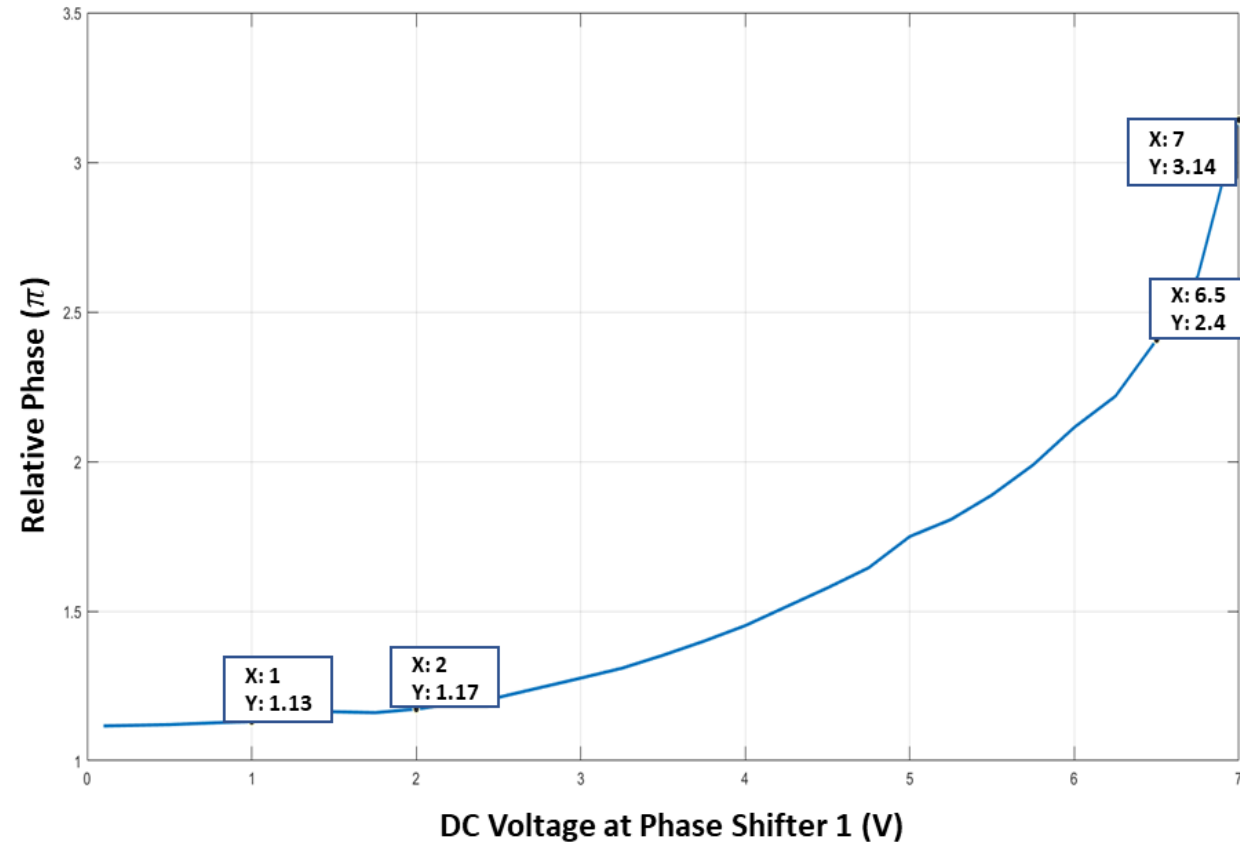
- I-V characterization for Heater resistance.
- $R1 = 560 \, \Omega$ & $R2 = 300 \, \Omega$
- Max Voltage to apply (Characterized) = 7V.



Mach-Zehnder Interferometer Phase Tuning



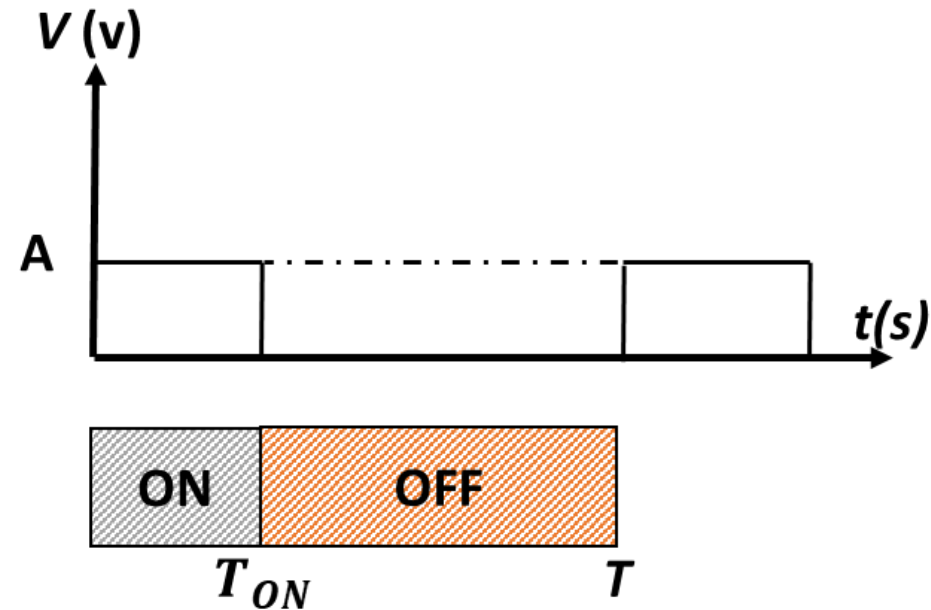
$$P \propto V_{DC}^2 / R \propto \Delta T \propto \Delta\phi$$



Quadratic response issues for phase differences $\Delta\phi$ at higher voltages!

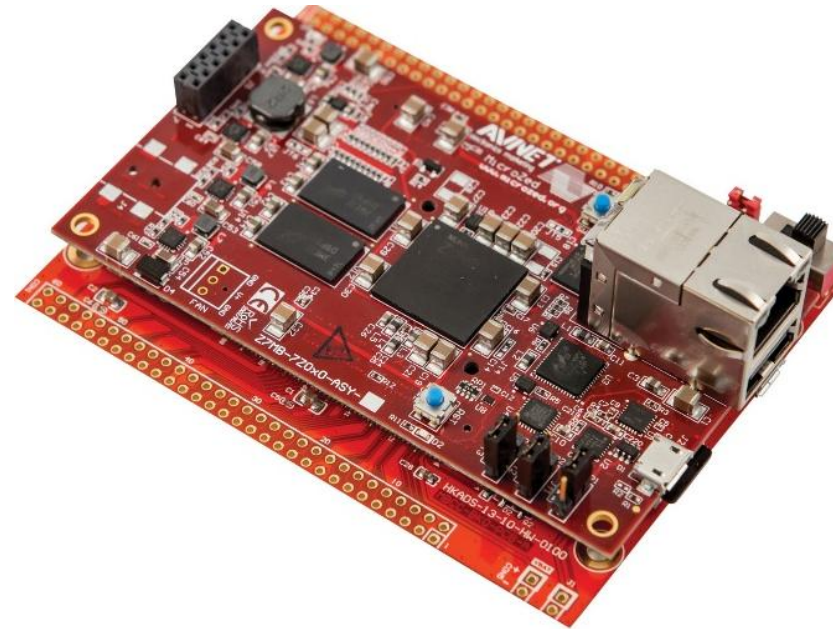
Solution? AC voltage (i.e., *PWM* signal)

Pulse Width Modulation signal (PWM)



$$P_{avg} = \frac{A^2}{R} * Duty\ Cycle\left(\frac{T_{ON}}{T}\right)$$

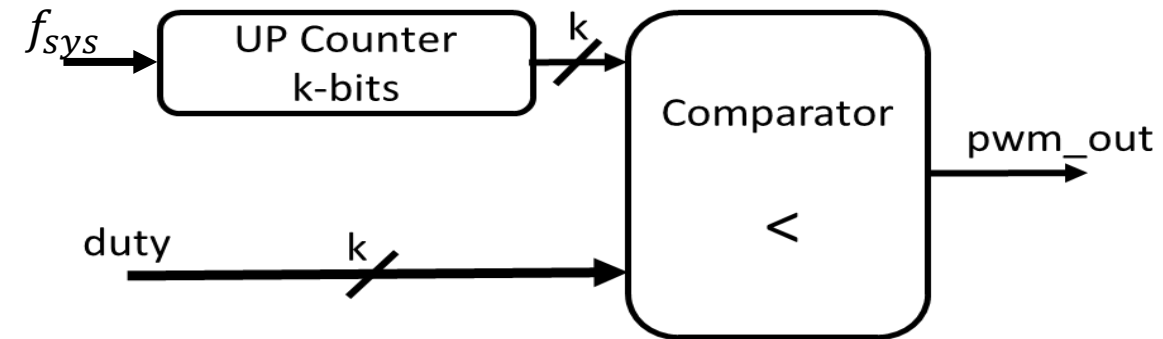
FPGA to generate PWM signal



- Field Programmable Gate Array (*FPGA*).
- FPGA used: Xilinx MicroZed (z7010) board.

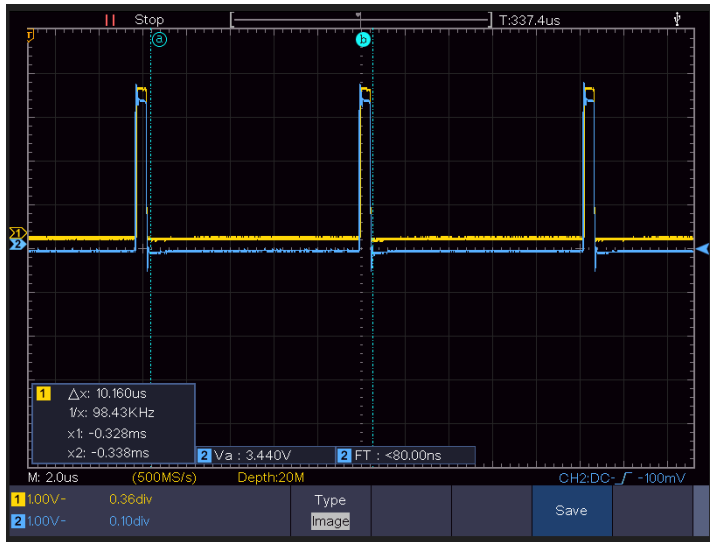
PWM Design Parameters

- Duty cycle: (0% -> 100 %)
- Resolution: (8-bits: $\frac{0}{256}, \frac{1}{256}, \dots, \frac{255}{256}, \frac{256}{256}$)

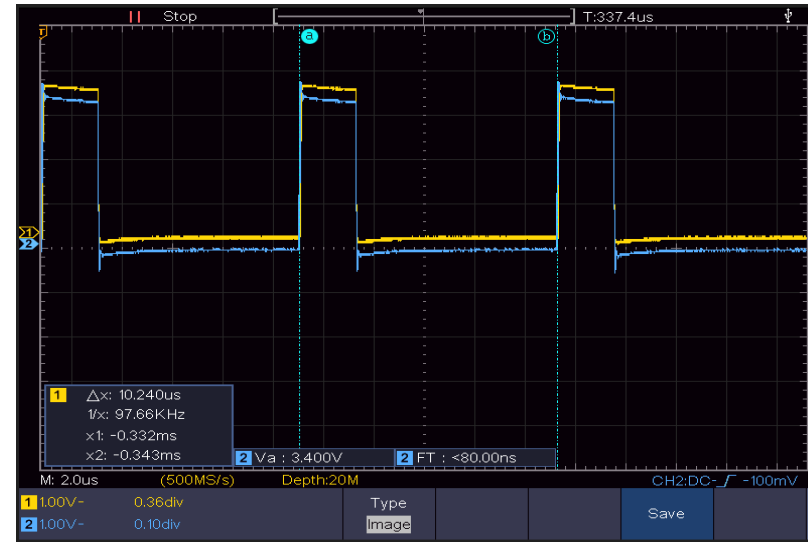


- Switching Frequency $f_{PWM} = \frac{1}{t_{PWM}} = \frac{f_{sys}}{2^k} = \frac{100 * 10^6}{2^8} = 390 \text{ KHz}.$

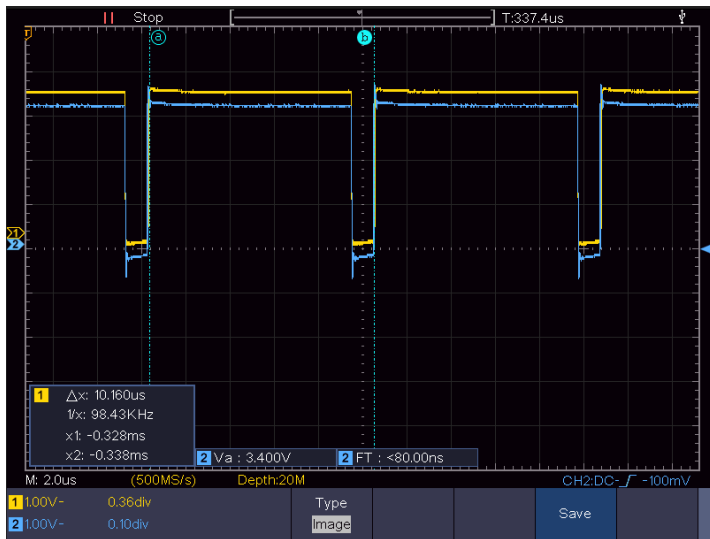
FPGA PWM Output signal (10-bits resolution)



*Outputs: 3.3V
Duty cycle=10%
Freq.= 98KHz*



*Outputs: 3.3V
Duty cycle=25%
Freq.= 98KHz*

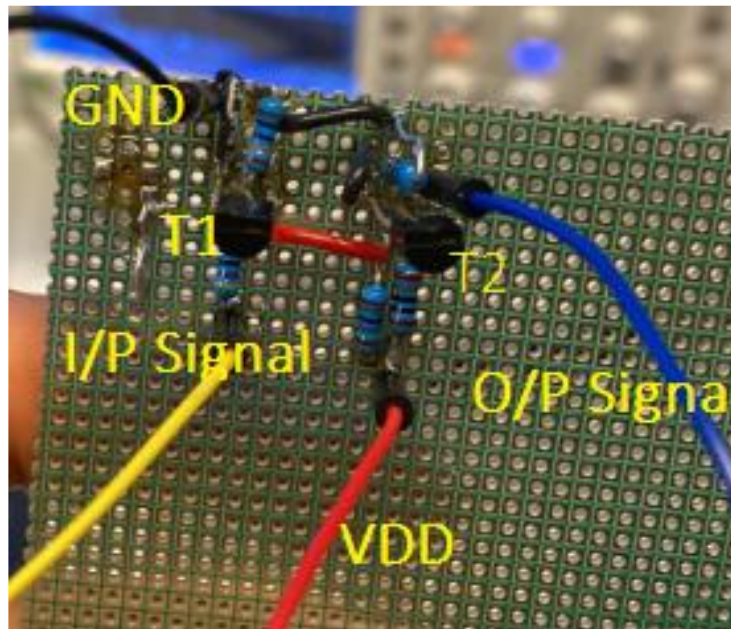
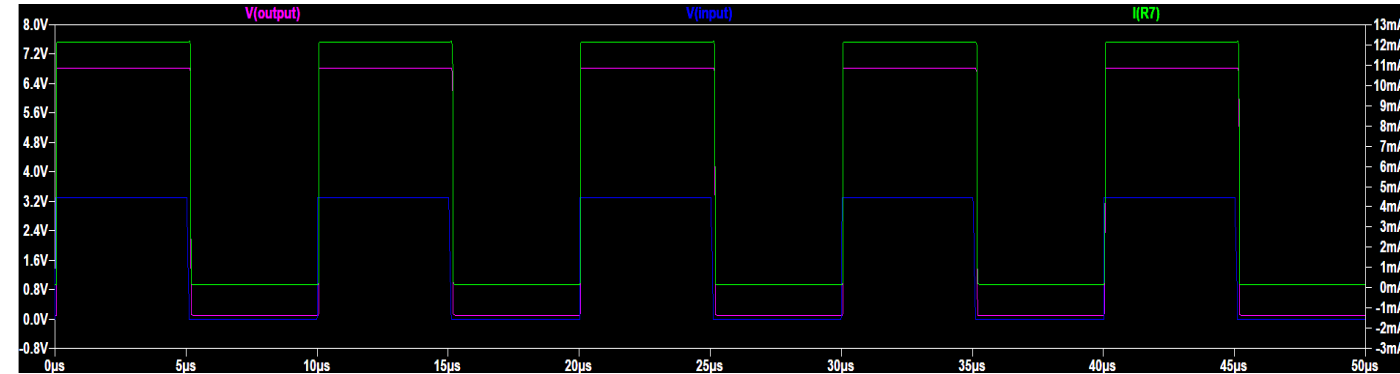
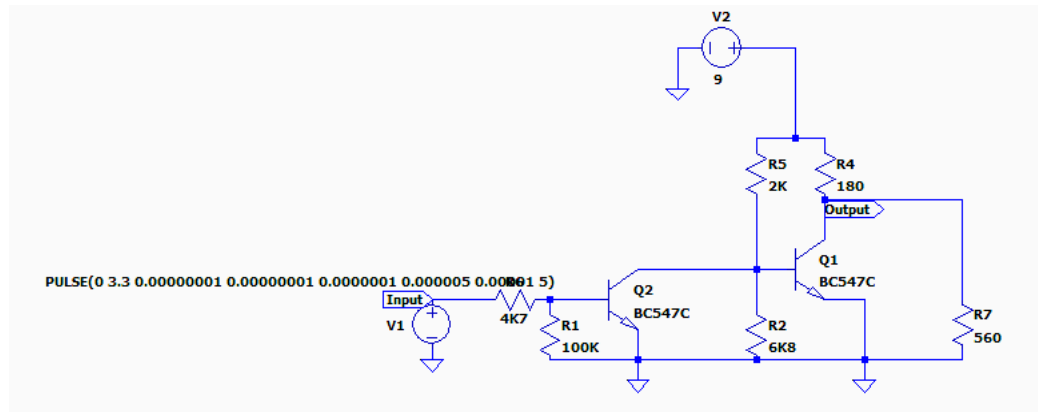


*Outputs: 3.3V
Duty cycle=90%
Freq.= 98KHz*

FPGA Outputs Voltage: 3.3V with no enough current for phase shifter heater.

Solution? Designing Booster Circuit

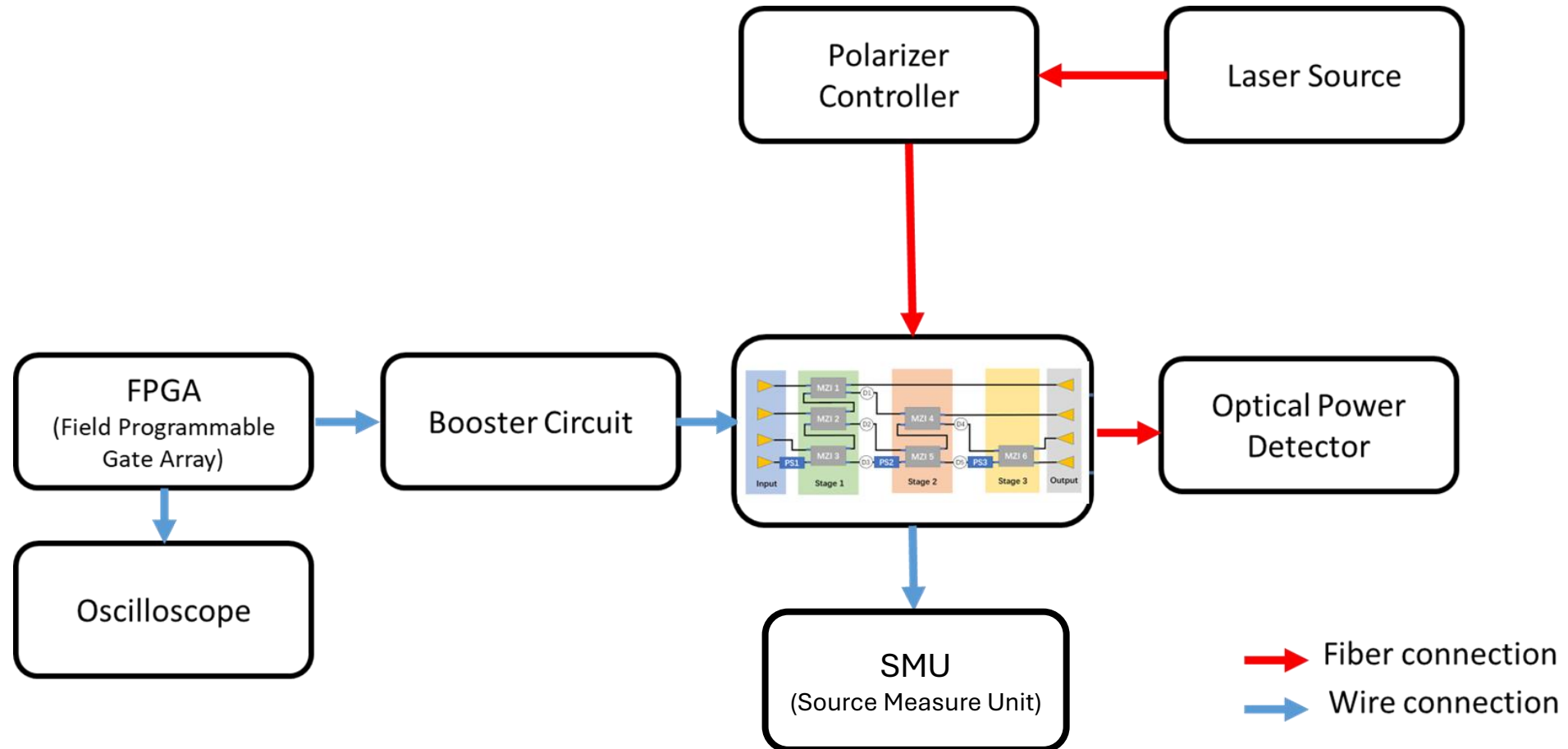
Boosting FPGA Outputs



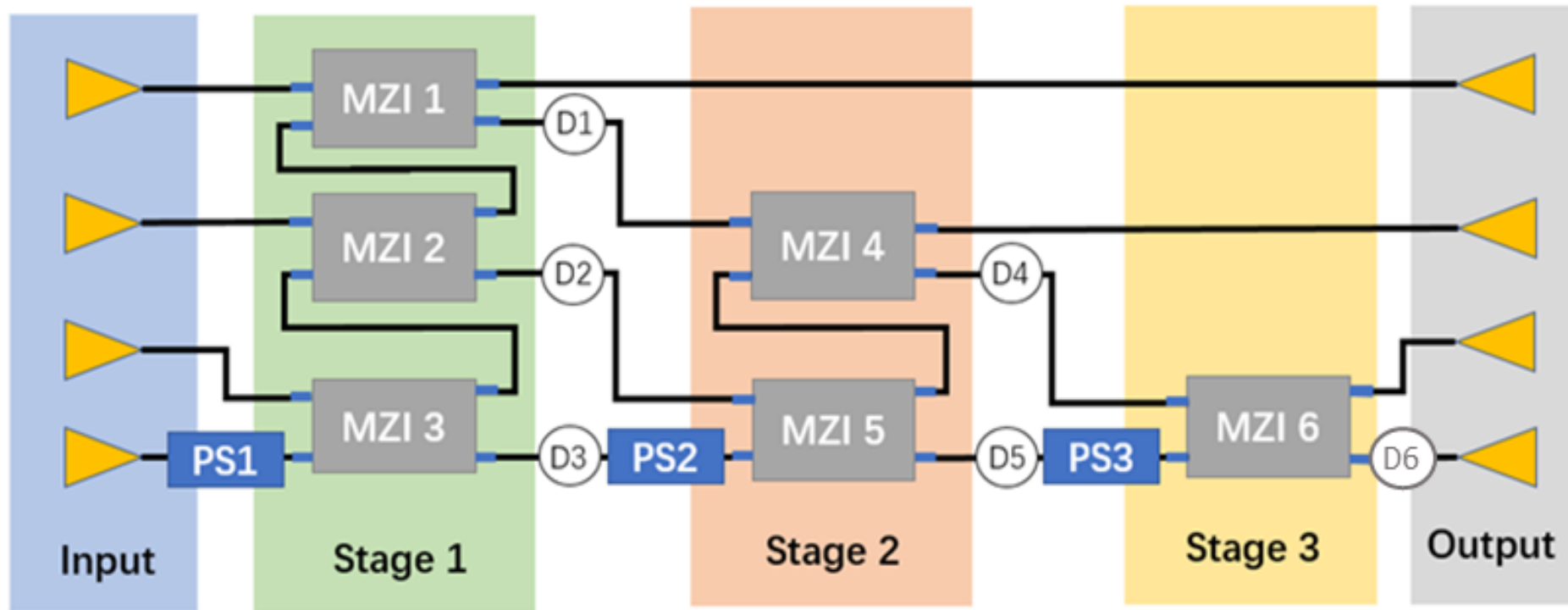
Maximum Switching Limitations



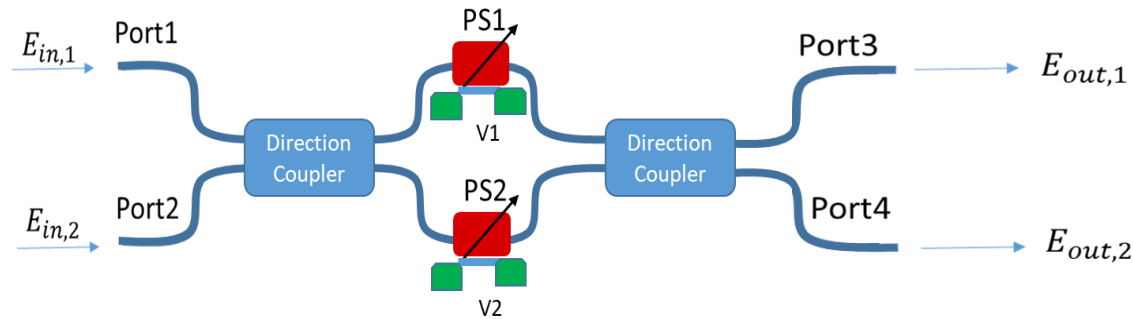
Digital Driving the Optical Phase Shift Heater



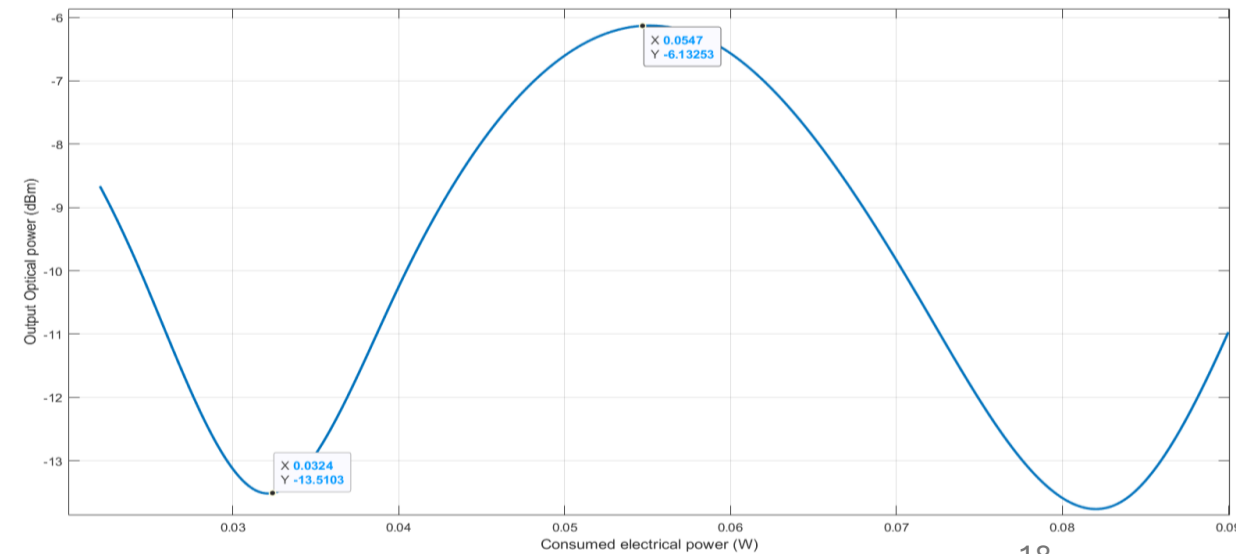
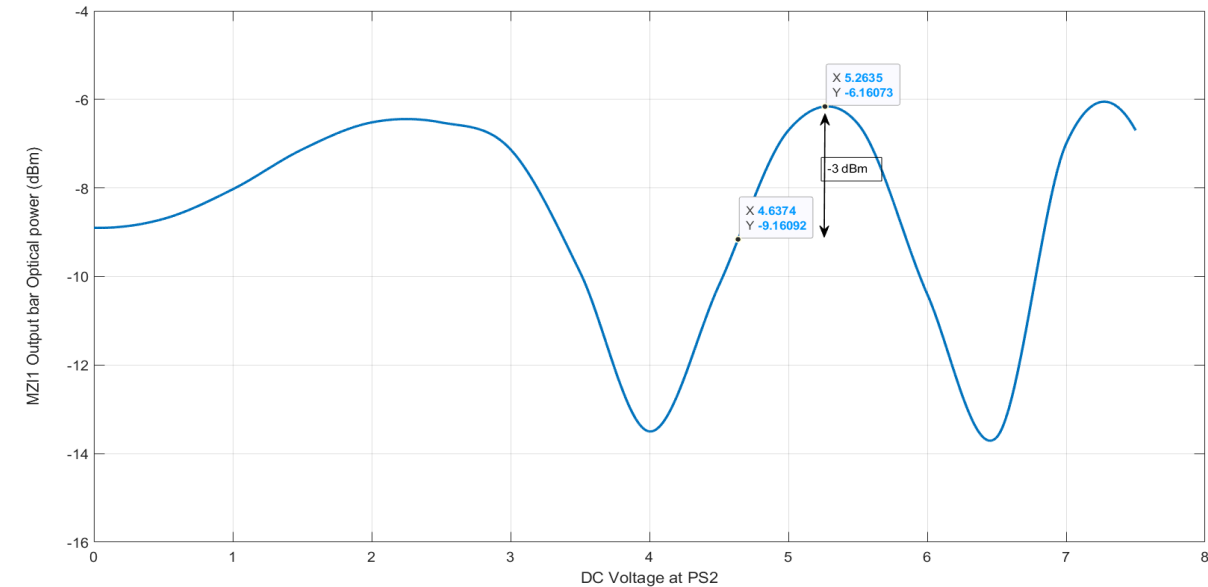
4x4 Optical Phase Shifter Circuit



Digital Driving for the Optical Phase Shift Heater



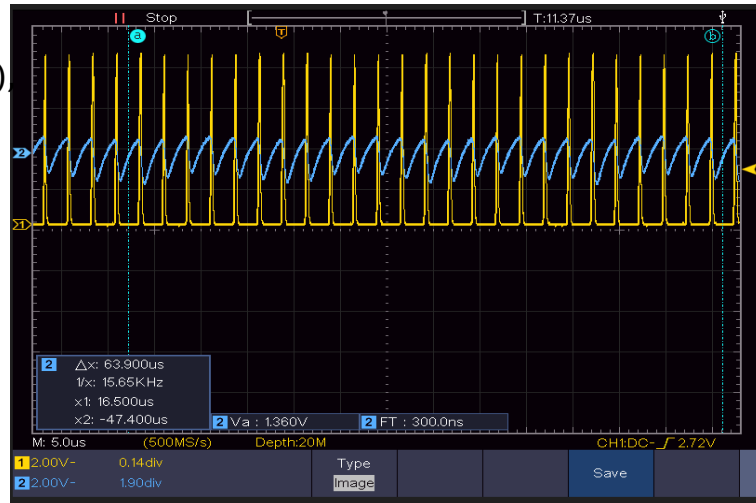
- MZI is fabricated in Cross State.
- MZI Operating voltage point @ -3dB: 4.6V
- $P_{\pi} = 34.4 \text{ mW}$



Digital Driving (FPGA) for Optical Phase Shift Heater

[Yellow]: PWM signal
 $f = 392$ KHz (8 bits resolution)
Duty Cycle: 5% .

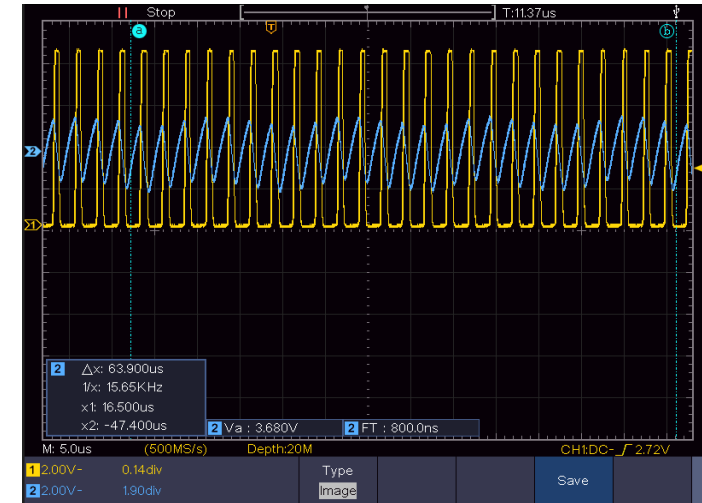
[Blue]: Photodetector Analog Output (relative phase)



[Yellow]: PWM signal

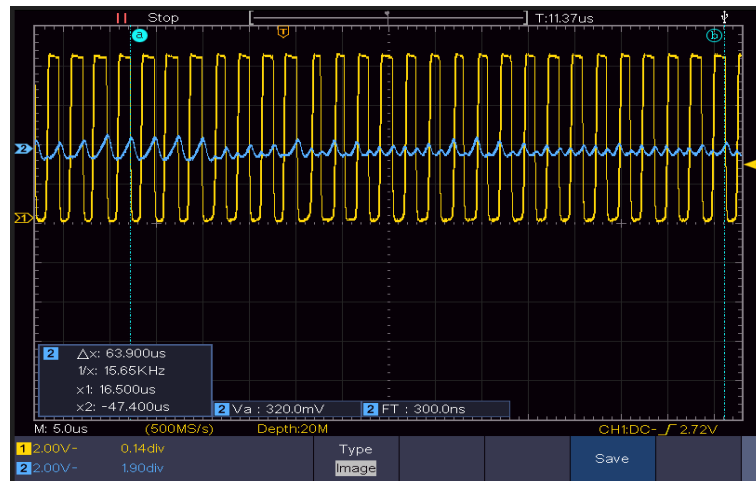
$f = 392$ KHz (8 bits resolution),
Duty Cycle: 10% .

[Blue]: Photodetector Analog Output (relative phase)



[Yellow]: PWM signal
 $f = 392$ KHz (8 bits resolution),
Duty Cycle: 50% .

[Blue]: Photodetector Analog Output (relative phase)



- Phase variation issue.
Solution? Higher frequency needed for no phase difference (steady).
- To test the MZI's heater limitation.
Action? Function Generator was used instead of FPGA

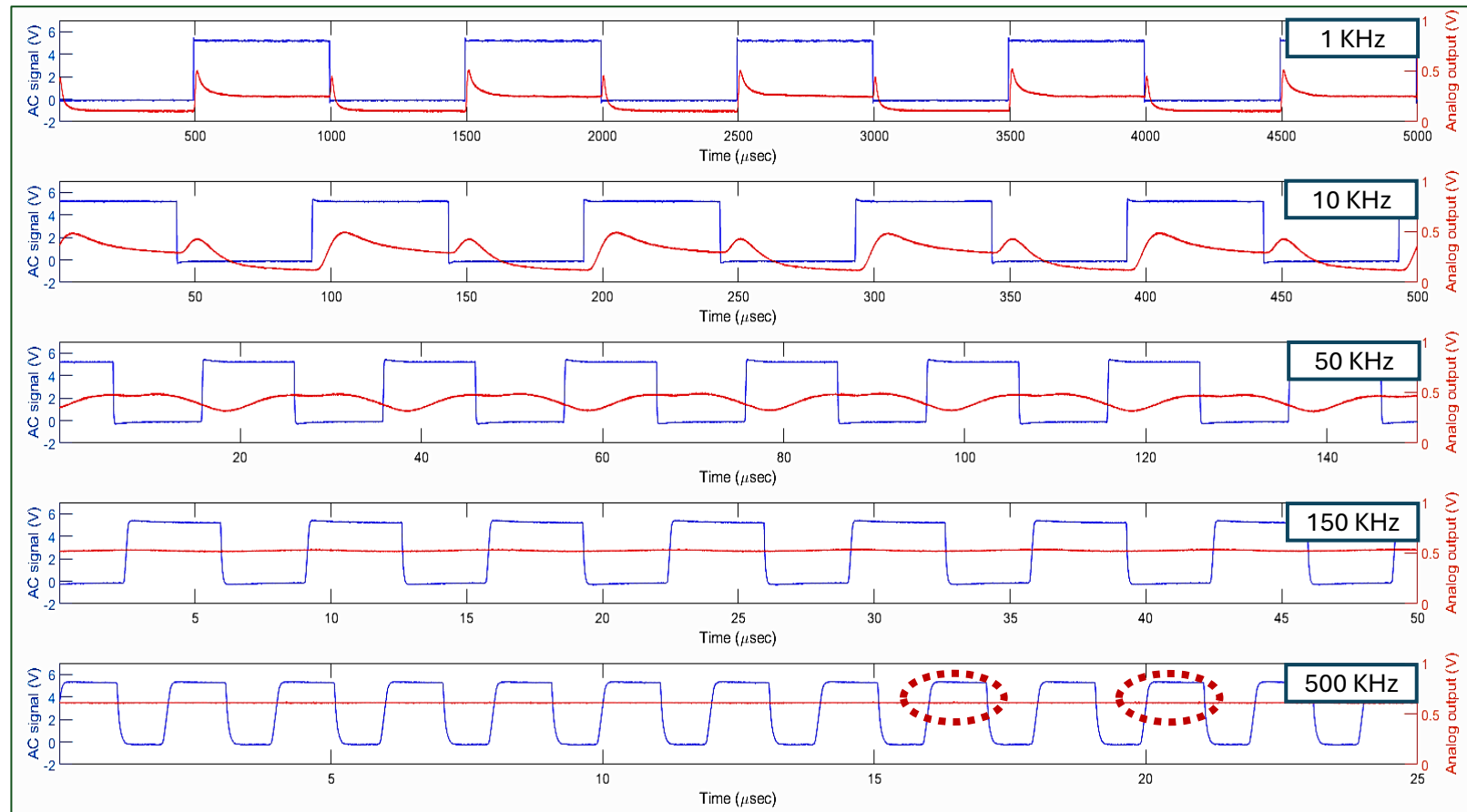
Digital Driving (Function Generator) for Optical Phase Shift Heater

- $f_{AC} = 1 \text{ KHz} \rightarrow 500 \text{ KHz}$
- *50% duty cycle max phase variation.*
- *Residual modulation at high frequencies, sampling precision issue.*

“Newport” meter Analog filter: 250 KHz (Datasheet)

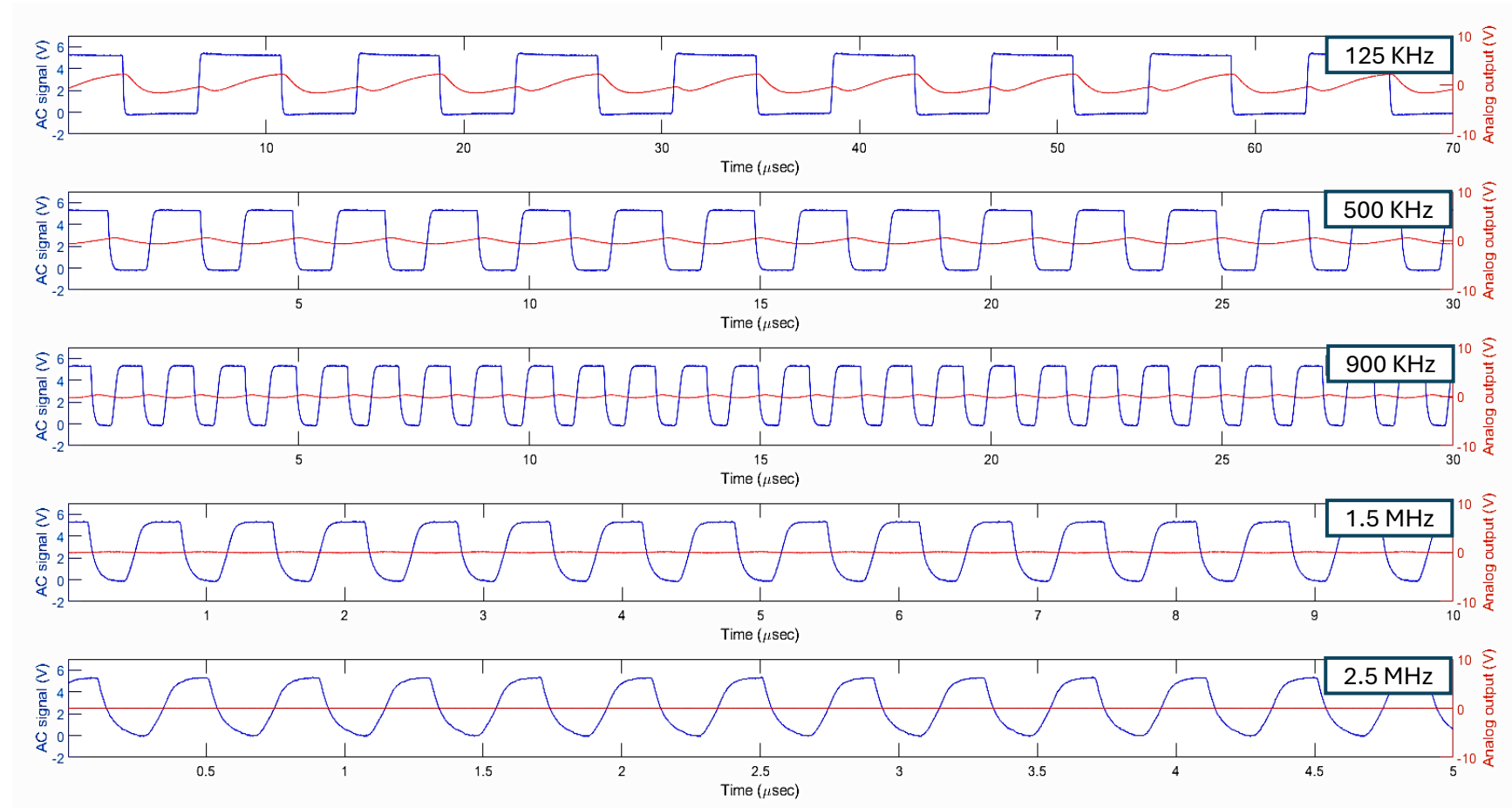
Solution?

RF High-speed photodetector instead of “Newport” photodetector power meter.



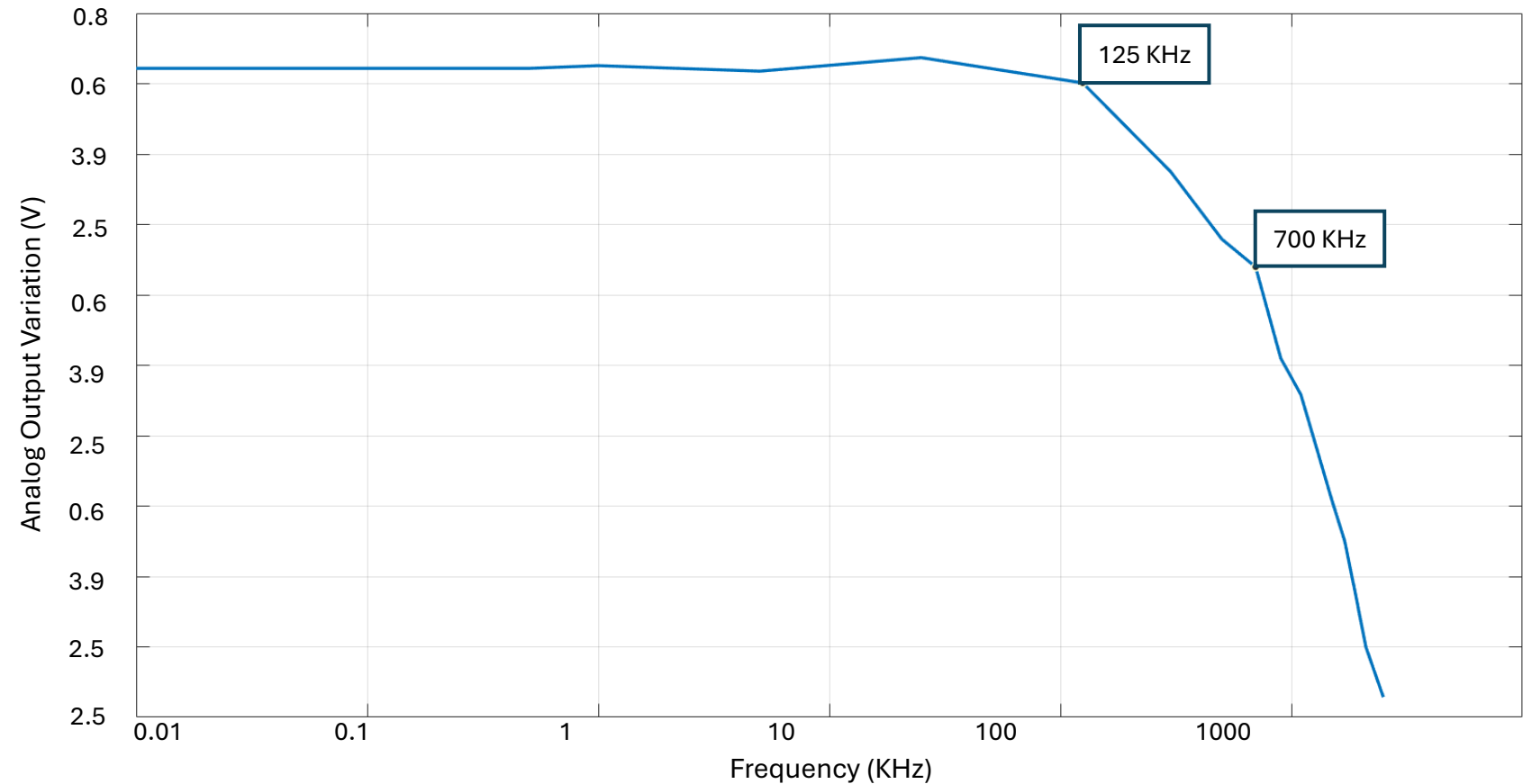
Digital driving (using Function Generator) for optical phase shift heater

- $f_{AC} = 125 \text{ KHz} \rightarrow 2.5 \text{ MHz}$.
- *Phase with no variation = 1.5 MHz.*



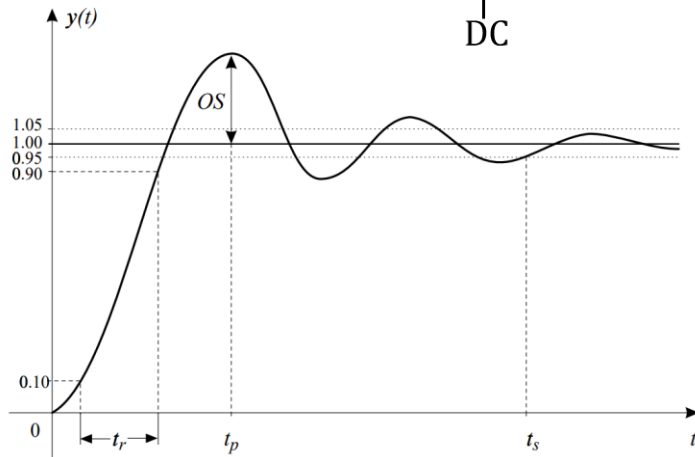
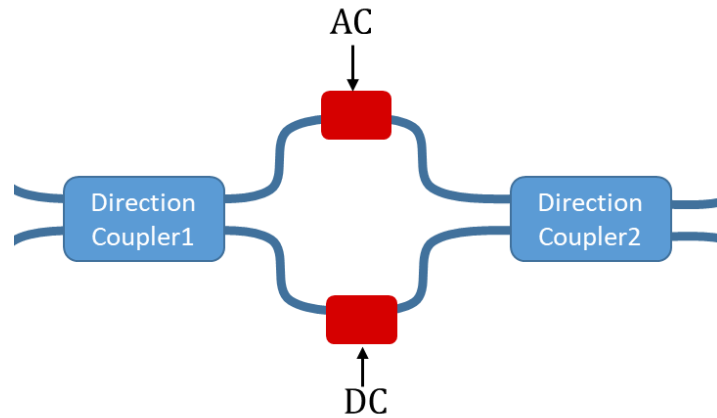
Digital driving for optical phase shift heater

- MZI f_{AC} response, log scale.
- $f_{AC,cutoff1} = 125 \text{ KHz}$
- $f_{AC,cutoff2} = 700 \text{ KHz}$

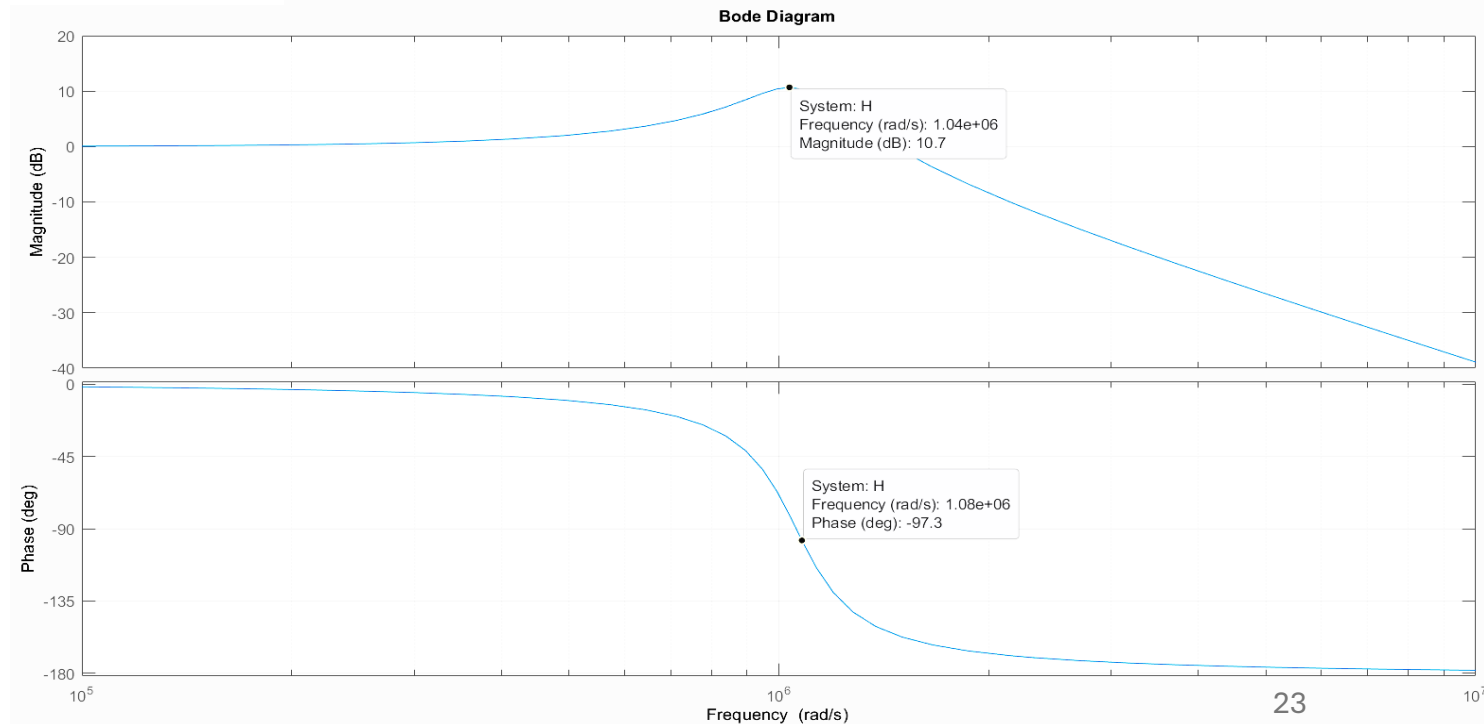
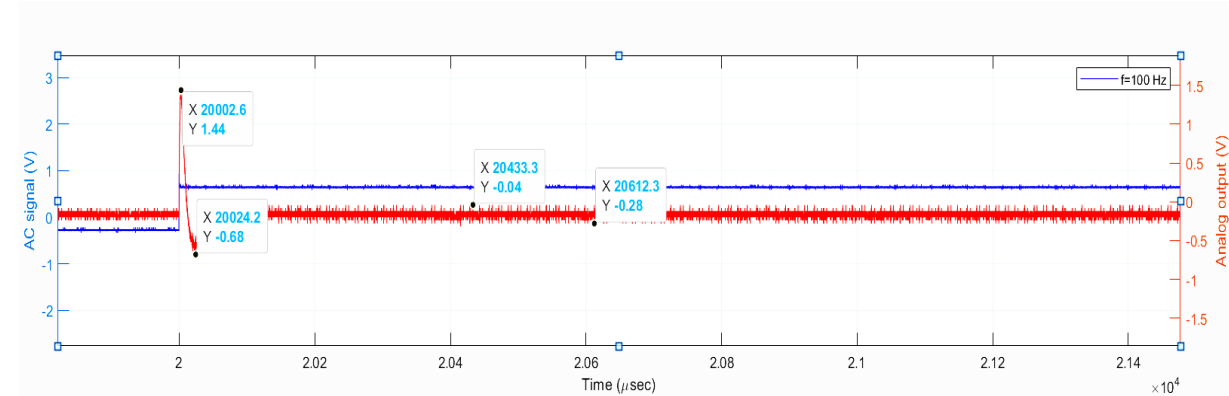


Digital driving for optical Phase Shift Heater

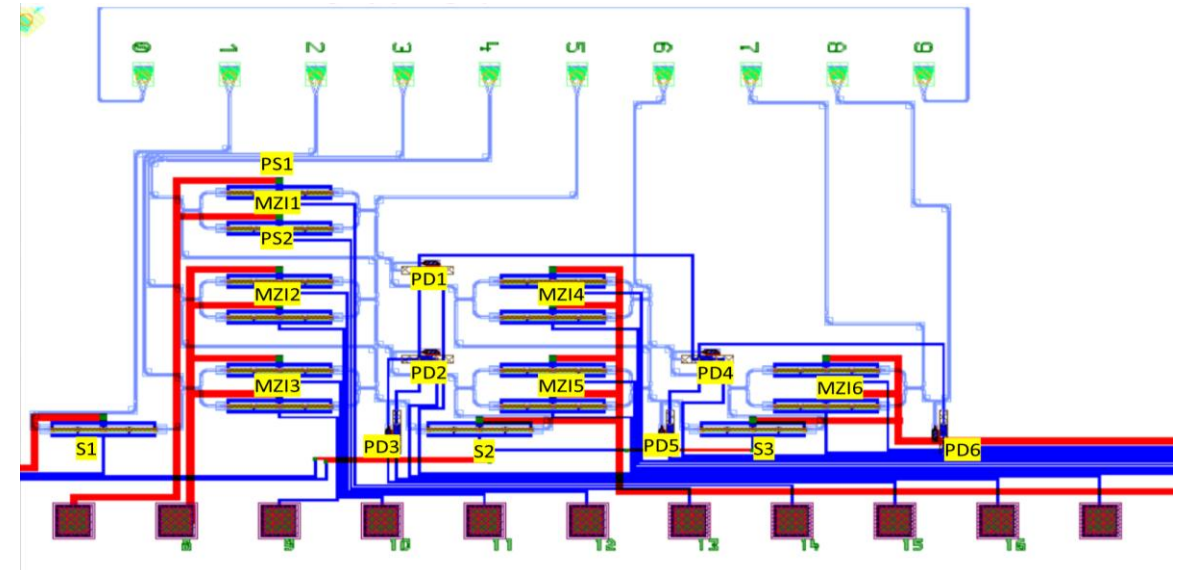
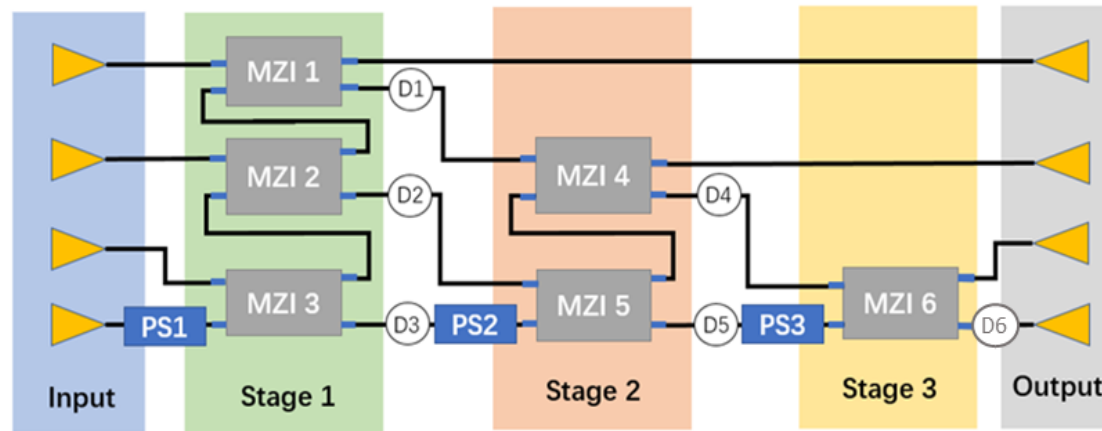
- MZI second order behavior with step response.



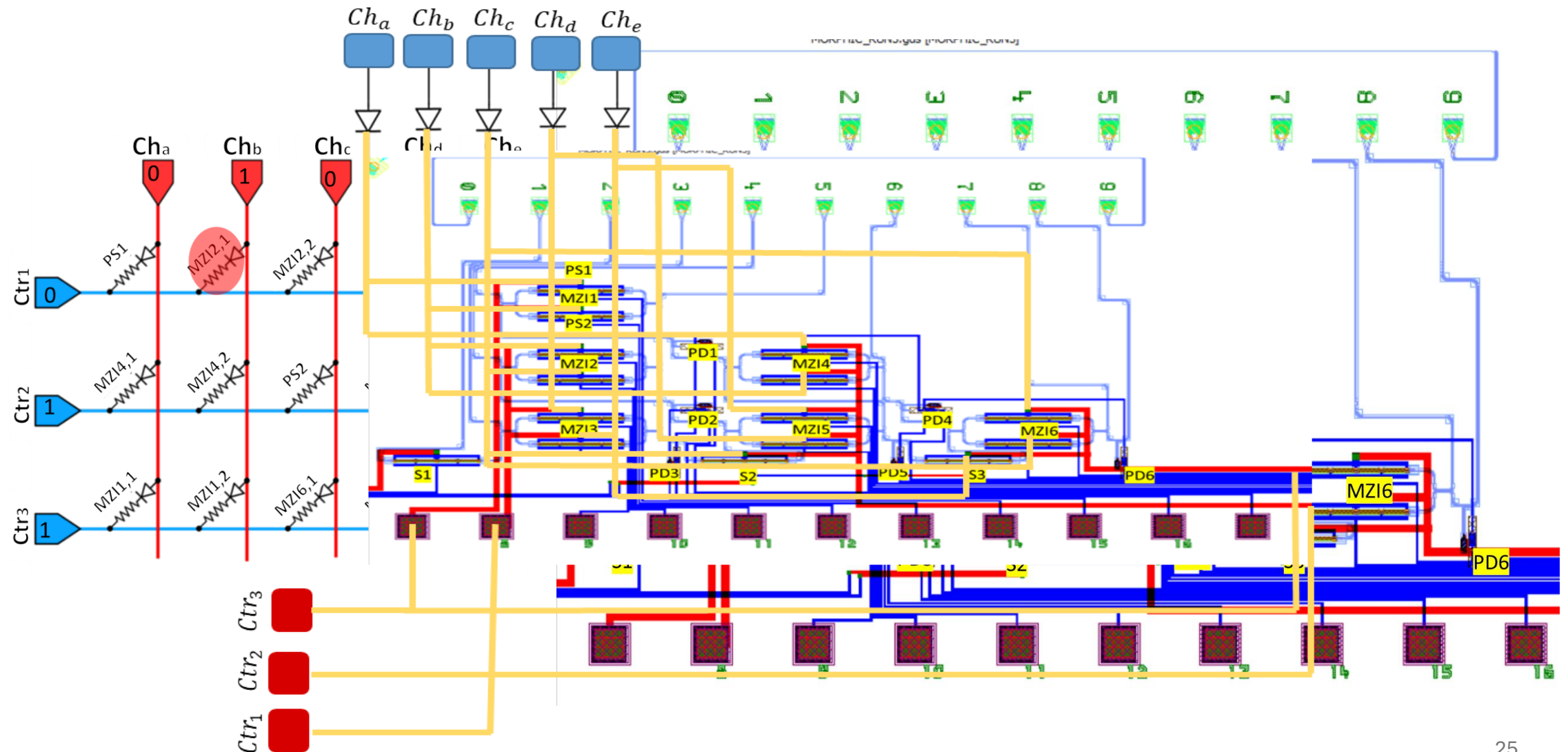
$$H(S) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1.121e12}{S^2 + 3.133e05 S + 1.121e12}$$



Matrix addressing configuration for the 4x4 Optical Phase Shifter Circuit



Matrix addressing configuration for the 4x4 circuit



Conclusion

My contributions

- PWM digital driving using *FPGA*.
- Linear phase variation with PWM signal.
- MZI characterization (Operating voltage point and P_π).
- Booster circuit design.
- MZI f_{AC} step response.
- Matrix addressing circuit design.

Future Work

- Advanced FPGA board for higher PWM frequency with more resolution.
- Thermal excitation model for RC circuit response of the phase shifter heater.
- Further study for the MZI step response.
- Control the matrix addressing with FPGA.



Thank You!

Any Questions?



